

# Lecture 2: Properties of the OLS estimator

Economics 527: Econometric Methods

Vadim Marmer, UBC

## The model and the assumptions

### Where the first lecture ended

- The first lecture produced the estimator and stopped there:

$$\hat{\beta} = (X^\top X)^{-1} X^\top Y = \left( \sum_{i=1}^n X_i X_i^\top \right)^{-1} \sum_{i=1}^n X_i Y_i,$$

the unique minimiser of  $\sum_{i=1}^n (Y_i - X_i^\top \beta)^2$ .

- Nothing so far says whether it is a good estimator.
- $\hat{\beta}$  is a function of the sample, so it is random, and in general it will not equal  $\beta$ .
- Three questions, taken in this order: whether  $\hat{\beta}$  is right on average, how far it moves from sample to sample, and whether a better estimator of the same kind exists.

### The model in matrix form

- Stack the  $n$  observations, as in the first lecture:

$$Y_{n \times 1} = \begin{pmatrix} Y_1 \\ \vdots \\ Y_n \end{pmatrix}, \quad X_{n \times k} = \begin{pmatrix} X_1^\top \\ \vdots \\ X_n^\top \end{pmatrix}, \quad U_{n \times 1} = \begin{pmatrix} U_1 \\ \vdots \\ U_n \end{pmatrix}.$$

- The  $n$  scalar equations  $Y_i = X_i^\top \beta + U_i$  become one:

$$Y_{n \times 1} = X_{n \times k} \beta_{k \times 1} + U_{n \times 1}.$$

- $\beta$  is fixed and unknown.  $X$ ,  $Y$  and  $U$  are random, and  $U$  is not observed.

### The four assumptions

- **Assumption 1.**  $Y = X\beta + U$  for some  $\beta \in \mathbb{R}^k$ .
- **Assumption 2.**  $E[U \mid X] = 0$ .
- **Assumption 3.**  $\text{Var}(U \mid X) = \sigma^2 I_n$  for some  $\sigma^2 > 0$ .
- **Assumption 4.**  $\text{rank}(X) = k$ .

## Unbiasedness

### What unbiased means

- $\hat{\beta}$  is a function of the sample, hence a random vector:

$$\hat{\beta}(\omega) = \left( \sum_{i=1}^n X_i(\omega) X_i(\omega)^\top \right)^{-1} \sum_{i=1}^n X_i(\omega) Y_i(\omega).$$

- With a continuously distributed error,  $P(\hat{\beta} = \beta) = 0$ . Hitting  $\beta$  exactly is not the standard.
- **Definition.** An estimator  $\hat{\theta}$  of  $\theta$  is **unbiased** if  $E[\hat{\theta}] = \theta$ , and **conditionally unbiased given  $X$**  if

$$E[\hat{\theta} | X] = \theta,$$

whatever the true value of  $\theta$ : the estimator is a rule that does not know  $\theta$ , and the equality must hold for every value  $\theta$  could take.

- The conditional statement is the stronger one, and it is the one this lecture proves.

## The sampling error

- Substitute Assumption 1 into the estimator; Assumption 4 is what makes  $(X^T X)^{-1}$  exist:

$$\begin{aligned}\hat{\beta} &= (X^T X)^{-1} X^T Y \\ &= (X^T X)^{-1} X^T (X\beta + U) \\ &= \underbrace{(X^T X)^{-1} X^T X}_{=I_k} \beta + (X^T X)^{-1} X^T U.\end{aligned}$$

- The first term produces  $\beta$ , and we have signal+noise decomposition:

$$\boxed{\hat{\beta} = \underbrace{\beta}_{\text{signal}} + \underbrace{(X^T X)^{-1} X^T U}_{\text{noise}}}$$

- Everything in this section and the next is a statement about the noise term.

## Taking expectations, first attempt

- Take expectations of the sampling error:

$$E[\hat{\beta}] = \beta + E[(X^T X)^{-1} X^T U].$$

- Suppose only that  $E[X_i U_i] = 0$  for every  $i$ . Then

$$E[X^T U] = E\left[\sum_{i=1}^n X_i U_i\right] = \sum_{i=1}^n E[X_i U_i] = 0.$$

- That is not enough, because in general

$$E[(X^T X)^{-1} X^T U] \text{ need not equal } E[(X^T X)^{-1}] E[X^T U].$$

- The two factors are built from the same  $X$ , and once  $E[U | X]$  is not zero, the expectation of their product need not factor.

## The expectation of a ratio

- The one-regressor case makes the failure visible:

$$E\left[\frac{\sum_i X_i U_i}{\sum_i X_i^2}\right] \text{ need not equal } \frac{E[\sum_i X_i U_i]}{E[\sum_i X_i^2]} = 0.$$

- In general the expectation of a ratio is not the ratio of expectations. For any two scalar random variables  $Z$  and  $S$ ,

$$E\left[\frac{Z}{S}\right] = \iint \frac{z}{s} f_{Z,S}(z, s) dz ds \text{ need not equal } \frac{\int z f_Z(z) dz}{\int s f_S(s) ds} = \frac{E[Z]}{E[S]}.$$

- The way out is to condition on  $X$ , which turns  $(X^T X)^{-1} X^T$  into a known matrix.

## Conditional unbiasedness: the claim

- **Claim.** Under Assumptions 1, 2 and 4,

$$\mathbb{E} [\hat{\beta} | X] = \beta.$$

- Assumption 3 is not used. Unbiasedness does not depend on the variance of the errors.

## Proof of conditional unbiasedness

- Condition the sampling error on  $X$ :

$$\begin{aligned}\mathbb{E} [\hat{\beta} | X] &= \mathbb{E} [\beta + (X^\top X)^{-1} X^\top U | X] \\ &= \beta + \mathbb{E} [(X^\top X)^{-1} X^\top U | X] \\ &= \beta + (X^\top X)^{-1} X^\top \underbrace{\mathbb{E} [U | X]}_{=0 \text{ by Assumption 2}} \\ &= \beta.\end{aligned}$$

- The third line takes out what is known, as in the first lecture. Given  $X$ , the matrix  $(X^\top X)^{-1} X^\top$  is known and comes out of the expectation.

## From conditional to unconditional

- The law of iterated expectations, from the first lecture, finishes the proof:

$$\mathbb{E} [\hat{\beta}] = \mathbb{E} [\mathbb{E} [\hat{\beta} | X]] = \mathbb{E} [\beta] = \beta.$$

- Conditional unbiasedness implies unbiasedness whenever the unconditional mean exists. The converse does not hold.

## A weaker condition

- Replace Assumption 2 by

$$\mathbb{E} [X_i U_i] = 0 \quad \text{for all } i,$$

which, together with  $\mathbb{E} [U_i] = 0$ , says that each regressor is uncorrelated with the error.

- $\beta$  is still identified under this condition.
- It is not enough for unbiasedness:

$$\mathbb{E} [\hat{\beta}] = \beta + \underbrace{\mathbb{E} [(X^\top X)^{-1} X^\top U]}_{\text{need not be 0}}.$$

- So  $\hat{\beta}$  need not be unbiased. The noise term is a nonlinear function of  $X$  and  $U$ , and once  $\mathbb{E} [U | X]$  is not zero, nothing forces its mean to vanish.

## The variance of a random vector

### The definition

- For a random vector  $V$  that is  $n \times 1$  with finite second moments, and with  $\mathbb{E} [V] = (\mathbb{E} [V_1], \dots, \mathbb{E} [V_n])^\top$ ,

$$\text{Var} (V) = \mathbb{E} \left[ \begin{pmatrix} V - \mathbb{E} [V] \\ \phantom{V - \mathbb{E} [V]} \phantom{V - \mathbb{E} [V]} \end{pmatrix} \begin{pmatrix} V - \mathbb{E} [V] \\ \phantom{V - \mathbb{E} [V]} \phantom{V - \mathbb{E} [V]} \end{pmatrix}^\top \right] \quad (n \times n).$$

- The product inside is an outer product, not an inner product, so the result is a matrix rather than a number.

## The matrix entry by entry

- Entry  $(i, j)$  is  $E[(V_i - E[V_i])(V_j - E[V_j])]$ , so

$$\text{Var}(V) = \begin{pmatrix} \text{Var}(V_1) & \text{Cov}(V_1, V_2) & \cdots & \text{Cov}(V_1, V_n) \\ \text{Cov}(V_2, V_1) & \text{Var}(V_2) & & \vdots \\ \vdots & & \ddots & \\ \text{Cov}(V_n, V_1) & \cdots & & \text{Var}(V_n) \end{pmatrix}.$$

- The variances sit on the diagonal and the covariances off it.

## Symmetry

- $\text{Var}(V)^\top = \text{Var}(V)$ , because  $\text{Cov}(V_i, V_j) = \text{Cov}(V_j, V_i)$ .
- The same statement from the definition, using that the transpose passes through the expectation, and then  $(AB)^\top = B^\top A^\top$ :

$$\text{Var}(V)^\top = E\left[\left((V - E[V])(V - E[V])^\top\right)^\top\right] = E\left[(V - E[V])(V - E[V])^\top\right] = \text{Var}(V).$$

## Positive semidefiniteness

- Claim.**  $\text{Var}(V)$  is positive semidefinite.
- Proof.** Take any non-random  $a \in \mathbb{R}^n$  and set  $W = a^\top(V - E[V])$ , a scalar. Then

$$\begin{aligned} a^\top \text{Var}(V) a &= a^\top E\left[(V - E[V])(V - E[V])^\top\right] a \\ &= E\left[\underbrace{a^\top(V - E[V])}_W \underbrace{(V - E[V])^\top a}_W\right] \\ &= E[W^2] \geq 0. \end{aligned}$$

- The constant  $a$  moves inside the expectation because expectation is linear.
- This is the argument the first lecture used for  $E[X_i X_i^\top]$ , with  $W_i = X_i^\top c$ .

## When it is positive definite

- For  $a \neq 0$ , with  $W = a^\top(V - E[V])$  as above, track the equality case:

$$a^\top \text{Var}(V) a = 0 \iff E[W^2] = 0 \iff a^\top(V - E[V]) = 0,$$

that is,

$$a_1(V_1 - E[V_1]) + \cdots + a_n(V_n - E[V_n]) = 0.$$

- So  $\text{Var}(V)$  is positive definite exactly when no non-zero linear combination of  $V_1, \dots, V_n$  is constant, that is, when no exact linear relation holds among the centred variables  $V_i - E[V_i]$ .

## Variance under a linear map

- Claim.** For a non-random  $m \times n$  matrix  $\Gamma$ ,

$$\boxed{\text{Var}(\Gamma V) = \Gamma \text{Var}(V) \Gamma^\top} \quad (m \times m).$$

- Proof.**  $E[\Gamma V] = \Gamma E[V]$ , so

$$\begin{aligned} \text{Var}(\Gamma V) &= E\left[(\Gamma V - \Gamma E[V])(\Gamma V - \Gamma E[V])^\top\right] \\ &= E\left[\Gamma(V - E[V])(V - E[V])^\top \Gamma^\top\right] \\ &= \Gamma E\left[(V - E[V])(V - E[V])^\top\right] \Gamma^\top = \Gamma \text{Var}(V) \Gamma^\top, \end{aligned}$$

using  $(AB)^\top = B^\top A^\top$  on the second factor, and linearity to move the non-random  $\Gamma$  and  $\Gamma^\top$  outside the expectation.

- Check against the scalar case: with  $m = n = 1$  this is  $\text{Var}(\gamma V) = \gamma^2 \text{Var}(V)$ .

### Adding a constant

- For a non-random  $\alpha$  that is  $m \times 1$ ,

$$\text{Var}(\alpha + \Gamma V) = \text{Var}(\Gamma V) = \Gamma \text{Var}(V) \Gamma^\top.$$

- The constant shifts the mean, so the deviations from the mean are unchanged and it drops out.
- This is why  $\beta$  will drop out of the variance of the sampling error.

### The conditional version

- The same definition with conditional expectations throughout:

$$\text{Var}(V | X) = \text{E} \left[ (V - \text{E}[V | X]) (V - \text{E}[V | X])^\top | X \right].$$

- Everything above holds conditionally, with  $\Gamma$  allowed to be a function of  $X$ . Given  $X$ , such a  $\Gamma$  is known and comes out of the conditional expectation, by taking out what is known, as in the proof of unbiasedness.
- When the conditional mean is zero, the definition simplifies. Under Assumption 2,

$$\text{Var}(U | X) = \text{E}[UU^\top | X].$$

## The variance of the OLS estimator

### The variance matrix of $\hat{\beta}$

- $\hat{\beta}$  is  $k \times 1$ , so  $\text{Var}(\hat{\beta} | X)$  is  $k \times k$ : the conditional variance of each estimated coefficient on the diagonal, the conditional covariances between them off it.
- Conditioning on  $X$  again, for the same reason it worked for unbiasedness: it makes  $(X^\top X)^{-1} X^\top$  a known matrix.

### The sandwich formula

- Start from the sampling error and name the matrix that multiplies  $U$ :

$$\hat{\beta} = \beta + \underbrace{(X^\top X)^{-1} X^\top}_{\Gamma_X} U.$$

- $\beta$  is a constant and drops out, and  $\Gamma_X$  is known given  $X$ , so the rule for a linear map applies:

$$\begin{aligned} \text{Var}(\hat{\beta} | X) &= \text{Var}(\beta + \Gamma_X U | X) \\ &= \text{Var}(\Gamma_X U | X) \\ &= \Gamma_X \text{Var}(U | X) \Gamma_X^\top. \end{aligned}$$

### The sandwich formula written out

- Substituting  $\Gamma_X = (X^\top X)^{-1} X^\top$ , with  $\Gamma_X^\top = X(X^\top X)^{-1}$  because  $X^\top X$  and its inverse are symmetric:

$$\boxed{\text{Var}(\hat{\beta} | X) = (X^\top X)^{-1} X^\top \text{Var}(U | X) X (X^\top X)^{-1}}$$

- The box uses Assumptions 1 and 4, and needs the errors to have finite conditional second moments, so that  $\text{Var}(U | X)$  exists. Assumption 2 enters only to write the middle factor as  $\text{E}[UU^\top | X]$ .
- Apart from requiring  $\text{Var}(U | X)$  to exist, nothing yet has been assumed about the errors' conditional variances or covariances.

### Assumption 3: homoskedasticity and no autocorrelation

- **Assumption 3.**  $\text{Var}(U | X) = \sigma^2 I_n$  for some  $\sigma^2 > 0$ , that is,

$$\text{Var}(U | X) = \begin{pmatrix} \sigma^2 & & 0 \\ & \ddots & \\ 0 & & \sigma^2 \end{pmatrix}.$$

- Read the diagonal: every error has the same conditional variance,

$$\text{Var}(U_1 | X) = \dots = \text{Var}(U_n | X) = \sigma^2.$$

- This is **homoskedasticity**. It rules out errors that are more variable at some values of the regressors than at others.
- Read the off-diagonal entries: for  $i \neq j$ , with the first equality by Assumption 2,

$$\text{Cov}(U_i, U_j | X) = \text{E}[U_i U_j | X] = 0.$$

- The errors are conditionally uncorrelated across observations. This is **no autocorrelation**, also called no serial correlation.
- Assumption 3 is these two statements together, and nothing more. It does not say the errors are independent, and it does not say they are normal.

### The variance of OLS under Assumption 3

- Put  $\text{Var}(U | X) = \sigma^2 I_n$  into the sandwich:

$$\begin{aligned} \text{Var}(\hat{\beta} | X) &= (X^\top X)^{-1} X^\top \sigma^2 I_n X (X^\top X)^{-1} \\ &= \sigma^2 (X^\top X)^{-1} \underbrace{X^\top X (X^\top X)^{-1}}_{= I_k} \\ &= \sigma^2 (X^\top X)^{-1}. \end{aligned}$$

- Under Assumptions 1 to 4:

|                                                                                                   |
|---------------------------------------------------------------------------------------------------|
| $\text{E}[\hat{\beta}   X] = \beta, \quad \text{Var}(\hat{\beta}   X) = \sigma^2 (X^\top X)^{-1}$ |
|---------------------------------------------------------------------------------------------------|

- This pair is the central result of the lecture.

### What $\sigma^2$ is

- From Assumptions 2 and 3 together,

$$\sigma^2 = \text{Var}(U_i | X) = \text{E}[U_i^2 | X],$$

and the law of iterated expectations gives

$$\sigma^2 = \text{E}[U_i^2] = \text{Var}(U_i),$$

the last step because  $\text{E}[U_i] = \text{E}[\text{E}[U_i | X]] = 0$  by Assumption 2.

- $\sigma^2$  is one unknown number, the same for every observation. Estimating it is a question for the next lecture.

## One regressor

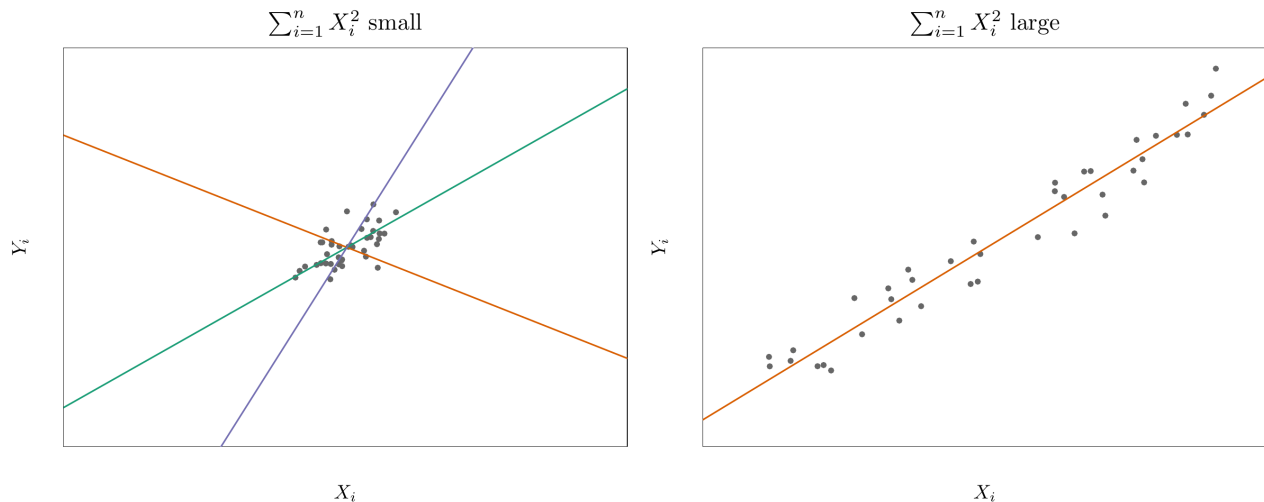
- Take  $k = 1$  and no intercept, so  $Y_i = \beta X_i + U_i$  and  $X$  is  $n \times 1$ . Then  $X^\top X = \sum_i X_i^2$  and

$$\text{Var}(\hat{\beta} | X) = \frac{\sigma^2}{\sum_{i=1}^n X_i^2}.$$

- Two things move the variance. Noisier errors raise it. A larger  $\sum_i X_i^2$  lowers it: with no intercept this is the sum of squares of the regressor about zero. With an intercept the same formula gives the variance of the slope, with  $\sum_i (X_i - \bar{X})^2$ , the spread of the regressor about its mean, in place of  $\sum_i X_i^2$ .

## Why spread in the regressor helps

- With the regressor bunched together, many quite different slopes fit the data about equally well, so the estimate moves a lot from sample to sample.
- With the regressor spread out, the slope is pinned down.



## Without homoskedasticity

- If the errors are heteroskedastic but still conditionally uncorrelated across observations, so that  $\text{Var}(U | X)$  is diagonal with entries that may depend on  $X$ ,

$$\text{Var}(U | X) = \Omega = \begin{pmatrix} \sigma_1^2 & & 0 \\ & \ddots & \\ 0 & & \sigma_n^2 \end{pmatrix},$$

the sandwich does not collapse. The middle factor becomes

$$X^\top \Omega X = \begin{pmatrix} X_1 & \cdots & X_n \end{pmatrix} \Omega \begin{pmatrix} X_1^\top \\ \vdots \\ X_n^\top \end{pmatrix} = \sum_{i=1}^n \sigma_i^2 X_i X_i^\top,$$

so

$$\text{Var}(\hat{\beta} | X) = (X^\top X)^{-1} \left( \sum_{i=1}^n \sigma_i^2 X_i X_i^\top \right) (X^\top X)^{-1}.$$

- Unbiasedness is untouched. Only the variance formula changes.

## Where each assumption was used

| Assumptions                             | What they deliver                                                                                                      |
|-----------------------------------------|------------------------------------------------------------------------------------------------------------------------|
| 1, 2, 4                                 | $E[\hat{\beta}   X] = \beta$ , and $E[\hat{\beta}] = \beta$ whenever that mean exists                                  |
| 1, 4, finite conditional second moments | $\text{Var}(\hat{\beta}   X) = \Gamma_X \text{Var}(U   X) \Gamma_X^\top$ , with<br>$\Gamma_X = (X^\top X)^{-1} X^\top$ |
| 1, 3, 4                                 | $\text{Var}(\hat{\beta}   X) = \sigma^2 (X^\top X)^{-1}$                                                               |

## The Gauss-Markov theorem

### Linear estimators

- **Definition.** An estimator of  $\beta$  is **linear** if it can be written

$$\tilde{\beta} = AY$$

for a matrix  $A$  that is  $k \times n$  and depends on  $X$  alone, not on  $Y$ .

- OLS is linear, with  $A = (X^\top X)^{-1} X^\top$ .
- Given  $X$ , each entry of a linear estimator is a fixed linear combination of the observations on the dependent variable.

### Two more linear estimators

- **The two-point slope.** With  $k = 1$ ,  $Y_i = \beta X_i + U_i$  and  $X_1 \neq X_n$ ,

$$\tilde{\beta} = \frac{Y_n - Y_1}{X_n - X_1} = -\frac{1}{X_n - X_1} Y_1 + \frac{1}{X_n - X_1} Y_n.$$

It uses two observations and throws the rest away.

- **Weighted least squares.** For a known  $n \times n$  matrix  $W$  that is symmetric and positive definite, and that may depend on  $X$  but not on  $Y$ ,

$$\tilde{\beta} = (X^\top W X)^{-1} X^\top W Y.$$

OLS is the case  $W = I_n$ . With  $W$  diagonal, this estimator weights observation  $i$  by the  $i$ th diagonal entry  $w_i$ , as in the first lecture's slide on other linear estimators. The inverse  $(X^\top W X)^{-1}$  exists: for  $c \neq 0$ ,  $Xc \neq 0$  by Assumption 4, so  $c^\top X^\top W X c = (Xc)^\top W (Xc) > 0$ .

### Both are unbiased

- The two-point slope, substituting  $Y_i = \beta X_i + U_i$  and using Assumption 2:

$$\tilde{\beta} = \beta + \frac{U_n - U_1}{X_n - X_1}, \quad E[\tilde{\beta} | X] = \beta.$$

- Weighted least squares, substituting  $Y = X\beta + U$  and using Assumption 2:

$$\tilde{\beta} = \beta + (X^\top W X)^{-1} X^\top W U, \quad E[\tilde{\beta} | X] = \beta.$$

- Being unbiased does not single OLS out. When  $n > k$ , there are many conditionally unbiased linear estimators.

## The variance of the weighted estimator

- The route that gave the sandwich formula applies, with  $A = (X^\top W X)^{-1} X^\top W$  in place of  $\Gamma_X$ . Since  $W^\top = W$ , the matrix  $X^\top W X$  and its inverse are symmetric too, so  $A^\top = W X (X^\top W X)^{-1}$ . Under Assumptions 1, 3 and 4,

$$\text{Var}(\tilde{\beta} | X) = \sigma^2 (X^\top W X)^{-1} X^\top W W X (X^\top W X)^{-1},$$

which reduces to  $\sigma^2 (X^\top X)^{-1}$  when  $W = I_n$ .

- Which of these matrices is smallest, and what smallest means for matrices, are the next questions.

## The comparison to settle

- For every linear conditionally unbiased  $\tilde{\beta}$ , compare the two conditional variance matrices:

$$\underset{k \times k}{\text{Var}(\tilde{\beta} | X)} - \underset{k \times k}{\text{Var}(\hat{\beta} | X)} \stackrel{?}{\geq} 0.$$

- Here  $\geq 0$  means positive semidefinite.

## The Gauss-Markov theorem

- Theorem.** Under Assumptions 1 to 4, let  $\tilde{\beta}$  satisfy
  - linearity:**  $\tilde{\beta} = AY$ , with  $A$  a  $k \times n$  matrix depending on  $X$  alone;
  - conditional unbiasedness:**  $E[\tilde{\beta} | X] = \beta$  for every value of  $\beta$ .
- Then

$$\text{Var}(\tilde{\beta} | X) - \text{Var}(\hat{\beta} | X) \text{ is positive semidefinite.}$$

- OLS is the **best linear unbiased estimator** (BLUE): best in the sense above, linear in  $Y$ , conditionally unbiased, and an estimator of  $\beta$ .
- The theorem is about conditional unbiasedness, because it compares conditional variances.

## Reading the matrix ordering

- If a  $k \times k$  matrix  $M$  is positive semidefinite, then  $c^\top M c \geq 0$  for **every**  $c \in \mathbb{R}^k$ .
- Take  $c$  to be the  $j$ th coordinate vector  $e_j$ , the vector with a 1 in position  $j$  and a 0 everywhere else:

$$e_j = (0, \dots, 0, 1, 0, \dots, 0)^\top, \quad \text{the 1 sitting in position } j.$$

- Writing the quadratic form out entry by entry, and using  $(e_j)_l = 0$  for every  $l \neq j$  and  $(e_j)_j = 1$ ,

$$e_j^\top M e_j = \sum_{l=1}^k \sum_{m=1}^k (e_j)_l M_{lm} (e_j)_m = M_{jj}.$$

- So every diagonal entry of a positive semidefinite matrix is non-negative.
- Apply this to  $M = \text{Var}(\tilde{\beta} | X) - \text{Var}(\hat{\beta} | X)$ , whose  $j$ th diagonal entry is the difference of the two variances of the  $j$ th coefficient:

$$\text{Var}(\tilde{\beta}_j | X) \geq \text{Var}(\hat{\beta}_j | X), \quad j = 1, \dots, k.$$

- A general  $c$  says the same about every linear combination  $c^\top \beta$  of the coefficients, since  $\text{Var}(c^\top V | X) = c^\top \text{Var}(V | X) c$ :

$$\text{Var}(c^\top \tilde{\beta} | X) \geq \text{Var}(c^\top \hat{\beta} | X) \quad \text{for every } c \in \mathbb{R}^k.$$

**Proof, step 1: linearity and unbiasedness force  $AX = I_k$**

- Take conditional expectations of  $\tilde{\beta} = AY$ :

$$\begin{aligned}\beta &= E[\tilde{\beta} \mid X] = E[A(X\beta + U) \mid X] \\ &= AX\beta + A \underbrace{E[U \mid X]}_{=0 \text{ by Assumption 2}} \\ &= AX\beta.\end{aligned}$$

- This must hold for every value of  $\beta$ , which is what unbiasedness means, so

$$\boxed{AX = I_k}$$

- Linearity plus conditional unbiasedness is exactly this one condition.

**Step 1 checked on the examples**

- OLS:  $A = (X^\top X)^{-1} X^\top$ , so  $AX = (X^\top X)^{-1} X^\top X = I_k$ .
- Weighted least squares:  $A = (X^\top W X)^{-1} X^\top W$ , so  $AX = (X^\top W X)^{-1} X^\top W X = I_k$ .
- The two-point slope, with  $k = 1$ :  $A$  has  $-1/(X_n - X_1)$  in position 1 and  $1/(X_n - X_1)$  in position  $n$ , so

$$AX = \frac{X_n - X_1}{X_n - X_1} = 1.$$

**Proof, step 2: the two sampling errors**

- Using  $AX = I_k$  for the general estimator, and the sampling error for OLS,

$$\hat{\beta} = \beta + (X^\top X)^{-1} X^\top U, \quad \tilde{\beta} = AX\beta + AU = \beta + AU.$$

- Both estimators differ from  $\beta$  only through a known matrix times  $U$ . Everything that follows is an expectation of  $UU^\top$ .

**The transpose rule for covariance matrices**

- For a  $k \times 1$  random vector  $V$  and an  $\ell \times 1$  random vector  $Z$ ,

$$\text{Cov}(V, Z) = E[(V - E[V])(Z - E[Z])^\top] \quad (k \times \ell),$$

and transposing gives

$$\text{Cov}(V, Z)^\top = E[(Z - E[Z])(V - E[V])^\top] = \text{Cov}(Z, V).$$

- The same definition and the same rule hold conditionally, with  $E[\cdot]$  replaced by  $E[\cdot \mid X]$  throughout. Step 4 needs this, because both orders appear there.

**Proof, step 3: the covariance equals the variance of OLS**

- Both estimators are conditionally unbiased, so their deviations from their conditional means are the two noise terms of step 2:

$$\begin{aligned}\text{Cov}(\tilde{\beta}, \hat{\beta} \mid X) &= E[(\tilde{\beta} - \beta)(\hat{\beta} - \beta)^\top \mid X] \\ &= E[AU U^\top X (X^\top X)^{-1} \mid X] \\ &= A \underbrace{E[UU^\top \mid X]}_{= \sigma^2 I_n \text{ by Assumptions 2 and 3}} X (X^\top X)^{-1} \\ &= \sigma^2 \underbrace{AX}_{= I_k} (X^\top X)^{-1} \\ &= \sigma^2 (X^\top X)^{-1}.\end{aligned}$$

- The right-hand side is  $\text{Var}(\hat{\beta} | X)$ , so

$$\boxed{\text{Cov}(\tilde{\beta}, \hat{\beta} | X) = \text{Var}(\hat{\beta} | X)}$$

- Every linear conditionally unbiased estimator has the same covariance with OLS, namely the variance of OLS itself. This is the one surprising step in the proof.

### Proof, step 4: the difference of the variances

- As for a scalar, expand the variance of the difference using the definition; the cross terms are the two covariance matrices:

$$\begin{aligned} \text{Var}(\tilde{\beta} - \hat{\beta} | X) &= \text{Var}(\tilde{\beta} | X) + \text{Var}(\hat{\beta} | X) \\ &\quad - \text{Cov}(\tilde{\beta}, \hat{\beta} | X) - \text{Cov}(\hat{\beta}, \tilde{\beta} | X). \end{aligned}$$

- Step 3 gives the first covariance term, the second is its transpose by the rule above, and  $\text{Var}(\hat{\beta} | X)$  is symmetric. All three terms in blue equal  $\text{Var}(\hat{\beta} | X)$ , and they combine into one with a minus sign:

$$\text{Var}(\tilde{\beta} - \hat{\beta} | X) = \text{Var}(\tilde{\beta} | X) - \text{Var}(\hat{\beta} | X).$$

- The left-hand side is a conditional variance matrix, hence positive semidefinite. Therefore

$$\text{Var}(\tilde{\beta} | X) - \text{Var}(\hat{\beta} | X) \geq 0,$$

which is the theorem.

### Uniqueness

- Suppose some linear conditionally unbiased  $\tilde{\beta}$  attains the same variance matrix. By the identity of step 4,

$$\text{Var}(\tilde{\beta} | X) = \text{Var}(\hat{\beta} | X) \implies \text{Var}(\tilde{\beta} - \hat{\beta} | X) = 0.$$

- A random vector with zero conditional variance equals its conditional mean (the equality case of positive semidefiniteness, one coordinate at a time), so

$$\tilde{\beta} - \hat{\beta} = d(X)$$

for some function of  $X$  alone.

- Both are conditionally unbiased, so taking conditional expectations,

$$\underbrace{\text{E}[\tilde{\beta} | X]}_{=\beta} = \underbrace{\text{E}[\hat{\beta} | X]}_{=\beta} + d(X) \implies d(X) = 0.$$

- Therefore  $\tilde{\beta} = \hat{\beta}$ . A linear conditionally unbiased competitor that is also best attains the same variance matrix: both  $M = \text{Var}(\tilde{\beta} | X) - \text{Var}(\hat{\beta} | X)$  and  $-M$  are then positive semidefinite, so  $c^\top M c = 0$  for every  $c$ , and a symmetric matrix with that property is zero. So OLS is not just a best linear unbiased estimator; it is the only one.

### Where Assumption 3 was used

- Assumption 3 entered in step 3, where  $\text{E}[UU^\top | X]$  was replaced by  $\sigma^2 I_n$ , and again when  $\sigma^2(X^\top X)^{-1}$  was recognised as  $\text{Var}(\hat{\beta} | X)$ . Without it the cancellation that produced  $\sigma^2 AX$  fails, and the covariance need not equal the variance of OLS.
- Assumptions 1, 2 and 4 give unbiasedness, and Assumptions 1 and 4 with finite conditional second moments give the sandwich formula. Assumption 3 is what makes OLS efficient.
- If Assumption 3 fails, OLS is still unbiased but need not be best: Assumption 3 is sufficient for efficiency, not necessary.

## Two extensions, and the summary

### Adding normality

- **Assumption 5.**  $U | X \sim N(0, \sigma^2 I_n)$ . Assumptions 1 to 5 define the classical normal regression model.
- Given  $X$ ,  $\hat{\beta} = \beta + (X^\top X)^{-1} X^\top U$  is a constant plus a linear function of  $U$ , and Assumption 5 fixes the conditional distribution of  $U$ . For non-random  $\alpha$  and  $\Gamma$ , such a function of a normal vector is normal:

$$V \sim N(\mu, \Sigma) \implies \alpha + \Gamma V \sim N(\alpha + \Gamma \mu, \Gamma \Sigma \Gamma^\top).$$

- Under Assumptions 1, 4 and 5, since Assumption 5 implies Assumptions 2 and 3, the conditional mean and variance are the ones already computed:

$$\boxed{\hat{\beta} | X \sim N(\beta, \sigma^2 (X^\top X)^{-1})}$$

- Nothing before this slide needed normality. It is what confidence intervals and tests will be built on.

### If Assumption 3 fails: generalised least squares

- Suppose now that  $\text{Var}(U | X) = \Omega$  for a known positive definite  $\Omega$ , not necessarily diagonal, with  $\Omega \neq \sigma^2 I_n$  for every  $\sigma^2 > 0$ . Let  $\Omega^{-1/2}$  be the symmetric positive definite matrix with  $\Omega^{-1/2} \Omega^{-1/2} = \Omega^{-1}$ , built from the eigenvalue decomposition  $\Omega = C \Lambda C^\top$ , and write  $\Omega^{1/2}$  for the inverse of  $\Omega^{-1/2}$ .
- Multiply the model through by  $\Omega^{-1/2}$ :

$$\underbrace{\Omega^{-1/2} Y}_{Y^*} = \underbrace{\Omega^{-1/2} X}_{X^*} \beta + \underbrace{\Omega^{-1/2} U}_{U^*}, \quad \mathbb{E}[U^* | X] = 0, \quad \text{Var}(U^* | X) = \Omega^{-1/2} \Omega \Omega^{-1/2} = I_n,$$

by taking out what is known and by the rule for a linear map.

- The transformed model satisfies all four assumptions ( $X^*$  has rank  $k$  because  $\Omega^{-1/2}$  is nonsingular), so Gauss-Markov applies to it. Since  $A \mapsto A \Omega^{1/2}$  matches the linear conditionally unbiased estimators of the two models one to one, OLS on the transformed model is also best among estimators linear in  $Y$  and conditionally unbiased, and it is

$$\hat{\beta}^* = (X^{*\top} X^*)^{-1} X^{*\top} Y^* = (X^\top \Omega^{-1} X)^{-1} X^\top \Omega^{-1} Y,$$

which is weighted least squares with  $W = \Omega^{-1}$ .

- In practice  $\Omega$  is not known. Later lectures return to this.

### Summary: unbiasedness and variance

- Substituting the model into the estimator gives the sampling error  $\hat{\beta} - \beta = (X^\top X)^{-1} X^\top U$ . Everything follows from it.
- **Unbiasedness.** Under Assumptions 1, 2 and 4, conditioning on  $X$  makes  $(X^\top X)^{-1} X^\top$  known, so  $\mathbb{E}[\hat{\beta} | X] = \beta$ , and the law of iterated expectations gives  $\mathbb{E}[\hat{\beta}] = \beta$  whenever that mean exists. The weaker condition  $\mathbb{E}[X_i U_i] = 0$  is not enough, because once  $\mathbb{E}[U | X]$  is not zero, the expectation of the noise term need not vanish, and the parabola shows that zero correlation does not make the conditional mean zero.
- **Variance.** Under Assumptions 1 and 4, given finite conditional second moments,

$$\text{Var}(\hat{\beta} | X) = (X^\top X)^{-1} X^\top \text{Var}(U | X) X (X^\top X)^{-1},$$

and adding Assumption 3, that  $\text{Var}(U | X) = \sigma^2 I_n$ , gives

$$\text{Var}(\hat{\beta} | X) = \sigma^2 (X^\top X)^{-1}.$$

With one regressor and no intercept this is  $\sigma^2 / \sum_i X_i^2$ : noisier errors raise it; a larger  $\sum_i X_i^2$  lowers it.

## Summary: efficiency

- **Gauss-Markov.** Under Assumptions 1 to 4, OLS has the smallest conditional variance among all estimators that are linear in  $Y$  and conditionally unbiased, whatever the value of  $\beta$ . For every such  $\tilde{\beta}$ , the difference of the two conditional variance matrices is positive semidefinite:

$$\text{Var}(\tilde{\beta} | X) - \text{Var}(\hat{\beta} | X) \geq 0,$$

so in particular  $\text{Var}(\tilde{\beta}_j | X) \geq \text{Var}(\hat{\beta}_j | X)$  for every coefficient.

- **The proof.** Linearity and conditional unbiasedness force  $AX = I_k$ . Assumption 3 then makes  $\text{Cov}(\tilde{\beta}, \hat{\beta} | X) = \text{Var}(\hat{\beta} | X)$ . Expanding  $\text{Var}(\tilde{\beta} - \hat{\beta} | X)$ , which is positive semidefinite, leaves exactly the difference of the two variances.
- **Uniqueness.** Equality of the variances forces  $\tilde{\beta} = \hat{\beta}$ .
- Assumption 3 is what buys efficiency. Assumptions 1, 2 and 4 give unbiasedness on their own.